Multivariable Calculus

Complete Handwritten Lecture Notes & Solved Problems

Part III: Multivariable Calculus

Functions of multiple variables, partial derivatives, total differentials, multiple integration, and vector calculus field equations.

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B5 Notebook

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Chapter 1: Multivariable Functions and Their Calculus

n-dimensional Euclidean space: $R^n = \{ (x_1, x_2, \ldots, x_n) \mid x_i \in R, i = 1, 2, \ldots, n \}$

For addition and scalar multiplication, R^n is a linear space over the real numbers.

Euclidean distance: $||X - Y||_n = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$: positivity, symmetry, triangle inequality

n-variable function: Let $\Omega \subset R^n$, if for any point $X \in \Omega$, there exists a unique value $u \in R^1$ corresponding to it, the corresponding rule is an n-variable function: $f: \Omega \subset R^n \to R^1, X \to u$.

Example: $u = \sqrt{1 - x^2 - y^2}$ $(x, y) \in D \subset R^2$

Example: Implicit representation: $F(x_1, x_2, \ldots, x_n, u) = 0$. (under certain conditions)

$R^n \to R^m$ vector-valued function: Let $\Omega \subset R^n$, if for any point $X \in \Omega$, there exists a vector $Y \in R^m$ corresponding to it, the corresponding rule is a vector-valued function from Ω to R^m. Each component y_j $(j = 1, 2, \ldots, m)$ of $Y = (y_1, y_2, \ldots, y_m)$ is an n-variable function: $y_j = f_j(x_1, x_2, \ldots, x_n), (x_1, \ldots, x_n) \in \Omega$.

Example: { $x = R \sin u \cos v$ $u \in [0, \pi]$ $y = R \sin u \sin v$ $v \in [0, 2\pi)$ $z = R \cos u$ $R^2 \to R^3$

Vector-valued function limit: $\lim_{X \to X_0} f(X) = A$ $(A \in R^m)$:

$\forall \varepsilon > 0, \exists \delta > 0, \forall X \in \Omega$ and $0 < ||X - X_0||_n < \delta \to ||f(X) - A||_m < \varepsilon$.

$\lim_{X \to X_0} f_j(X) = a_j, j = 1, 2, \ldots, m$.

Proof of limit non-existence: counterexample of different limits.

Proof of limit existence: use ε-δ through inequality proof.

Vector-valued function continuity: $f(X)$ is continuous at X_0 point. $\forall \varepsilon > 0, \exists \delta > 0, \forall X: ||X - X_0||_n < \delta, \to ||f(X) - f(X_0)||_m < \varepsilon$.

Theorem: Continuous vector-valued functions are closed under addition, subtraction, multiplication, and composition. Continuous vector-valued functions have the extreme value theorem and the intermediate value theorem.

kth order infinitesimal function: $p^k = ||x - x_0||^k$. If $\frac{f(x)}{p^k} = constant$ as $x \to x_0$, then $f(x)$ is of kth order.

e.g., $f(x) = \sum_{i,j=1}^{n} a_{ij} x_i x_j = O(p^2)$.

n-ary function's total differential: $u = f(x)$ is defined in the ball $B(x_0, r)$. For $x \in B(x_0, r)$, $\Delta u = f(x) - f(x_0) = a_1(x_1 - x_1^0) + \ldots + a_n(x_n - x_n^0) + o(p)$. Then u is differentiable at x_0, where $o(p)$ is the higher-order infinitesimal differential.

As $x - x_0 \to 0$, $f(x) \to f(x_0)$, so differentiable functions are continuous at x_0.

By the definition of total differential, if the n-ary function $u = f(x)$ is differentiable at x_0, then its partial derivatives $\frac{\partial u}{\partial x_1}(x_0), \ldots, \frac{\partial u}{\partial x_n}(x_0)$ all exist, and $a_i = \frac{\partial u}{\partial x_i}(x_0)$.

The converse of the above proposition is not true.

By the definition of total differential, if u has all partial derivatives at x_0, and $\Delta u - \frac{\partial u}{\partial x_1}\Delta x_1 - \ldots - \frac{\partial u}{\partial x_n}\Delta x_n$ is of higher-order infinitesimal p, then u is differentiable at x_0. Otherwise, it is not differentiable.

To determine differentiability, a sufficient condition is: if u has all partial derivatives at x_0 and they are continuous, then u is differentiable at x_0.

Where $x^0 = (cos\alpha_1, cos\alpha_2, \ldots, cos\alpha_n)$, if u is differentiable at x_0, then

$\frac{\partial u}{\partial t}|_{p^0} = \frac{\partial u}{\partial x_1}|_{p^0} cos\alpha_1 + \ldots + \frac{\partial u}{\partial x_n}|_{p^0} cos\alpha_n$ (differentiable implies directional derivative exists)

$\frac{\partial u}{\partial t}|_{p^0} = \frac{\partial u}{\partial x_1}|_{p^0} cos\alpha_1 + \ldots + \frac{\partial u}{\partial x_n}|_{p^0} cos\alpha_n$ (differentiable implies directional derivative exists)

The gradient of an n-variable function u at point X0 is the vector of partial derivatives. The direction of the maximum directional derivative is the unit vector in that direction. (If differentiable, it exists and is non-zero).

grad $u(X_0) = (\partial u/\partial x_1, \partial u/\partial x_2, \ldots, \partial u/\partial x_n) \to (x_1, \ldots, x_n)$: $R^n \to R^n$

Theorem: $u = f(x)$ in Ω has k-order partial derivatives (u is k-order differentiable in Ω) then u has r-order mixed partial derivatives ($2 \le r \le k$) and the order of differentiation is irrelevant.

The higher-order differential of an n-variable function: $d^k u = (\partial u/\partial x_1\, dx_1 + \ldots + \partial u/\partial x_n\, dx_n)^k = D^k u$

D: linear operator

Vector-valued function differentiation: $\Delta f = f(x_0 + \Delta x) - f(x_0) = A \Delta x + o(\|\Delta x\|)$. $A \Delta x$: $f(x)$ has a differential $d f(x_0)$; $o(\|\Delta x\|)$ is higher-order infinitesimal as $\|x - x_0\|$.

Vector-valued function is differentiable at x0 if and only if each component (n-variable) function is differentiable at x0. Among them

$A = J(f(x_0)) = (\partial f_1/\partial x_1, \ldots, \partial f_m/\partial x_n) = (\nabla f_1)|_{x_0}, \ldots, (\nabla f_m)|_{x_0}$.

$= (\partial(f_1, \ldots, f_m)/\partial(x_1, \ldots, x_n))|_{x_0}$.

Chain rule for vector-valued functions: If $u = g(x)$ is differentiable at x0, $y = f(u)$ at $u_0 = g(x_0)$ is differentiable, then $f \circ g$ is differentiable at x0. $d(f \circ g)(x_0) = J[f(u_0)] \cdot J[g(x_0)]$. That is

$J[f \circ g(x_0)] = J[f(u_0)] \cdot J[g(x_0)]$.

$i = 1, \ldots, n$.

Example: $y = f(u_1, u_2, \ldots, u_m)$, $u_j = g_j(x_1, \ldots, x_n)$, $j = 1, \ldots, m$.

The Existence of Implicit Functions (Multivariate or Vector-Valued Functions):

- For an n+1 variable function F(x1, x2, ..., xn, y) in a neighborhood B(x0, y0, r) of point (x0, y0), if F is k-times continuously differentiable and F(x0, y0) = 0, ∂F/∂y(x0, y0) ≠ 0, then there exists an n-variable implicit function y = f(x), such that F(x, f(x)) = 0. The function f is k-times continuously differentiable.

$$\frac{\partial y}{\partial x_i} = -\frac{\partial F}{\partial x_i}(x, y) \quad i = 1, 2, \ldots, n$$

$$\frac{\partial^2 y}{\partial x_i \partial x_j} = -\frac{\partial}{\partial x_j}\left[\frac{\partial F}{\partial x_i}(x, y)\right]$$

Where ∂F/∂xi(x, y(x)) is the composite function of ∂F/∂xi(x, y) and y = y(x).

Example: F(x, y, z) = 0, the implicit function z = z(x, y) is 2-times continuously differentiable. Then,

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = -\frac{\partial}{\partial y}\left[\frac{\partial F}{\partial x}(x, y, z(x, y))\right]$$

$$= -\frac{\partial}{\partial y}\left[\frac{\partial F}{\partial x}(x, y, z(x, y))\right] = -\frac{\partial}{\partial y}\left[\frac{\partial F}{\partial x}(x, y, z(x, y))\right]$$

$$\frac{\partial^2 z}{\partial y \partial x} = -\frac{\partial}{\partial y}\left[\frac{\partial F}{\partial x}(x, y, z(x, y))\right] = -\frac{\partial}{\partial y}\left[\frac{\partial F}{\partial x}(x, y, z(x, y))\right]$$

Where $\partial y[\partial F/\partial x(x, y, z(x, y))] = \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) +$

$\partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) + \partial^2F/\partial y\partial x(x, y) + \partial^2F/\partial z\partial x(x, y) + \partial^2F/\partial x^2(x, y) +$
∂^2F

The function is continuously differentiable.

$J(f(x)) = \partial(y1, ..., ym) / \partial(x1, ..., xn) = - (\partial(E1, ..., Em) / \partial(y1, ..., ym))^{-1} \partial(F1, ..., Fm) / \partial(x1, ..., xn)$.

Let Y = f(X) be a R^n → R^m continuously differentiable vector-valued function. If the Jacobian matrix J(f(X)) is invertible at X ∈ Ω, then f is invertible in B(X, δ). The inverse vector-valued function X = g(Y) is also continuously differentiable. $J(g(Y)) = [J(f(X))]^{-1}$

Example: A system of linear equations formed by full-rank matrices.

Surface's explicit representation: S = z = f(x, y), (x, y) ∈ Dxy ⊂ R^2.

Tangent plane: z - z0 = (∂f/∂x)(x0, y0)(x - x0) + (∂f/∂y)(x0, y0)(y - y0).

Normal vector: n = (∂f/∂x, ∂f/∂y, -1) |_p0.

Tangent plane equation: (∂f/∂x)(x0, y0) = (∂f/∂y)(x0, y0) = -1.

Surface's parametric representation: S = {x = x(u, v), y = y(u, v), z = z(u, v)}, (u, v) ∈ Duv ⊂ R^2.

Tangent plane: {x - x0 = (∂x/∂u)(u0, v0)(u - u0) + (∂x/∂v)(u0, v0)(v - v0),

y - y0 = (∂y/∂u)(u0, v0)(u - u0) + (∂y/∂v)(u0, v0)(v - v0),

z - z0 = (∂z/∂u)(u0, v0)(u - u0) + (∂z/∂v)(u0, v0)(v - v0)}.

Normal vector (all tangent lines are perpendicular): n = (∂(y, z) / ∂(u, v), ∂(z, x) / ∂(u, v), ∂(x, y) / ∂(u, v)) |_(u0, v0).

Tangent plane equation: (∂(y, z) / ∂(u, v)) |_(u0, v0) = (∂(z, x) / ∂(u, v)) |_(u0, v0) = (∂(x, y) / ∂(u, v)) |_(u0, v0).

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The surface's implicit function representation: S: F(x, y, z) = 0.

The tangent plane:

$$\left(\frac{\partial F}{\partial x}\right)_{P_0}(x-x_0)+\left(\frac{\partial F}{\partial y}\right)_{P_0}(y-y_0)+\left(\frac{\partial F}{\partial z}\right)_{P_0}(z-z_0)=0$$

The normal vector: n = $(\frac{\partial F}{\partial x},\frac{\partial F}{\partial y},\frac{\partial F}{\partial z})_{P_0}$

The normal equation:

$$\frac{x-x_0}{\frac{\partial F}{\partial x}}=\frac{y-y_0}{\frac{\partial F}{\partial y}}=\frac{z-z_0}{\frac{\partial F}{\partial z}}$$

The parametric representation of a space curve: L:

$$\begin{cases} x=x(t)\\ y=y(t)\\ z=z(t)\end{cases}$$

The tangent line:

$$\begin{cases} x-x_0=x'(t_0)(t-t_0)\\ y-y_0=y'(t_0)(t-t_0)\\ z-z_0=z'(t_0)(t-t_0)\end{cases}$$

The tangent plane:

$$(x'(t_0),y'(t_0),z'(t_0))\cdot((x-x_0),(y-y_0),(z-z_0))^T=0$$

The implicit function representation of the space curve:

$$\begin{cases} F_1(x,y,z)=0\\ F_2(x,y,z)=0\end{cases}$$

The tangent plane at P_0:

$$\left(\frac{\partial F_1}{\partial x}\right)_{P_0}(x-x_0)+\left(\frac{\partial F_1}{\partial y}\right)_{P_0}(y-y_0)+\left(\frac{\partial F_1}{\partial z}\right)_{P_0}(z-z_0)=0$$

$$\left(\frac{\partial F_2}{\partial x}\right)_{P_0}(x-x_0)+\left(\frac{\partial F_2}{\partial y}\right)_{P_0}(y-y_0)+\left(\frac{\partial F_2}{\partial z}\right)_{P_0}(z-z_0)=0$$

The intersection of the two tangent planes is the tangent line: T = $(\frac{\partial F_1}{\partial x}, \frac{\partial F_1}{\partial y}, \frac{\partial F_1}{\partial z})_{P_0} \times (\frac{\partial F_2}{\partial x}, \frac{\partial F_2}{\partial y}, \frac{\partial F_2}{\partial z})_{P_0}$

Using the tangent direction, the parametric equation of the tangent line can be written.

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The second-order Taylor formula with Peano remainder for an n-variable function:

$$f(\mathbf{x}) = f(\mathbf{x}_0) + J(f(\mathbf{x}_0))\Delta\mathbf{x} + \frac{1}{2!}(\Delta\mathbf{x})^T H(\mathbf{x}_0)\Delta\mathbf{x} + o(\|\Delta\mathbf{x}\|^2)$$

n-variable function

Necessary condition for a critical point: f is differentiable at x0 (in all variables), if x0 is a local maximum (minimum) of f, then

$$\frac{\partial f}{\partial x_i}(x_0) = 0 \Leftrightarrow \nabla f(x_0) = \vec{0}$$

If f is twice continuously differentiable in a neighborhood of x0, x0 is a critical point:

{If H(x0) is positive definite, x0 is a local minimum}

{If H(x0) is negative definite, x0 is a local maximum}

Example: Let u = f(x, y, z) = x^3 + y^2 + z^2 + 6xy + 2z. Find the critical points.

By

$$u_x = u_y = u_z = 0$$

we get two critical points P1(6, -18, -1), P2(0, 0, -1)

The Hessian at P1 is:

$$H(P_1) = \begin{bmatrix} 36 & 6 & 0 \\ 6 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The Hessian at P2 is:

$$H(P_2) = \begin{bmatrix} 0 & 6 & 0 \\ 6 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

∴ P1 is a local minimum, P2 is not a local minimum.

Conditional extremum problem:

$$\begin{cases} \min(\max) f(x_1, x_2, \ldots, x_n) \\ g(x_1, x_2, \ldots, x_n) = 0 \end{cases}$$

Objective function (n variables), constraint.

Theorem: The conditional extremum problem in Ω has an extremum point at Lagrange function

$$\mathcal{L}(x_1, x_2, \ldots, x_n, \lambda) = f(x_1, \ldots, x_n) + \lambda g(x_1, \ldots, x_n)$$

The extremum point of the Lagrange function is the extremum point of the original function.

Therefore, by solving the Lagrange function to find the extremum points and then substituting them into the objective function to compare the values, we can obtain the solution.

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Example

Space Ellipsoid

\[

\left\{

\begin{array}{l}

\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \

lx + my + nz = 0

\end{array}

\right.

\]

Objective Function

\[

\min(\max) \quad x^2 + y^2 + z^2

\]

Constraints:

\[

\begin{aligned}

&\text{Lagrange Function:} \quad L = x^2 + y^2 + z^2 + \lambda_1 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right) + \lambda_2 (lx + my + nz) \

&\text{First-order conditions:} \

&\quad L_x = 2x + \lambda_1 \frac{2x}{a^2} + \lambda_2 l = 0 \

&\quad L_y = 2y + \lambda_1 \frac{2y}{b^2} + \lambda_2 m = 0 \

&\quad L_z = 2z + \lambda_1 \frac{2z}{c^2} + \lambda_2 n = 0 \

&\text{Substitute the constraints:} \

&\quad a^* = \left(\frac{B + \sqrt{B^2 - AC}}{A} \right)^{\frac{1}{2}}, \quad b^* = \left(\frac{B - \sqrt{B^2 - AC}}{A} \right)^{\frac{1}{2}} \

&\text{Where:} \

&\quad A = a^2 l^2 + b^2 m^2 + c^2 n^2, \quad B = \frac{1}{2} \left[l^2 a^2 (b^2 + c^2) + m^2 b^2 (c^2 + a^2) + n^2 c^2 (a^2 + b^2) \right], \quad C = a^2 b^2 c^2

\end{aligned}

\]

Chapter 2: Variable Parameter Integrals

Definition: $I(y) = \int_{[a,b]} f(x,y)\, dx$, $y \in [c,d]$.

Continuity: $f(x,y)$ is continuous in $D = [a,b] \times [c,d]$, then $I(y)$ is continuous in $[c,d]$.

Ordering: $f(x,y)$ and $\partial f/\partial y(x,y)$ are continuous in $D = [a,b] \times [c,d]$, then $\forall y_0 \in (c,d)$

$\left[\partial/\partial y \int_{[a,b]} f(x,y)\, dx\right]_{y=y_0} = \int_{[a,b]} \left[\partial/\partial y\, f(x,y)\right]_{y=y_0} dx$.

Leibniz's Formula: $f(x,y)$ and $\partial f/\partial y(x,y)$ are continuous in $D = [a,b] \times [c,d]$, $\alpha(y)$ and $\beta(y)$ are in $[c,d]$ and bounded, then

$\partial/\partial y \int_{[\alpha(y),\beta(y)]} f(x,y)\, dx = \int_{[\alpha(y),\beta(y)]} \partial f/\partial y(x,y)\, dx + f(\beta(y),y)\, \beta'(y) - f(\alpha(y),y)\, \alpha'(y)$.

Ordering*: $f(x,y)$ is continuous in $D = [a,b] \times [c,d]$, then

$\int_{[c,d]} dy \int_{[a,b]} f(x,y)\, dx = \int_{[a,b]} dx \int_{[c,d]} f(x,y)\, dy$.

For improper integrals, add a uniform convergence condition to use the above theorems.

Example: Calculate $\int_{[0,\infty]} (e^{-ax} - e^{-bx}) / x\, dx$ $(b \geq a > 0)$.

By $\int_{[a,b]} e^{-xy}\, dy = (e^{-ax} - e^{-bx}) / x$, then the original integral $= \int_{[0,\infty]} dx \int_{[a,b]} e^{-xy}\, dy$.

When $a \leq y \leq b$, $|e^{-xy}| \leq e^{-ax}$, thus $\int_{[a,b]} e^{-xy}\, dx$ in $y \in [a,b]$ is uniformly convergent, so

the original integral $= \int_{[a,b]} dy \int_{[0,\infty]} e^{-xy}\, dx = \ln(b/a)$.

Chapter 3: Double Integrals

The double integral of a bounded function f(x1, ..., xn) over the bounded region Ω in Rn is:

$$\int_\Omega \cdots \int_\Omega f(x_1, \dots, x_n)\, dx_1 \dots dx_n$$

Theorem: If f is a continuous function on the closed region Ω, then f has a Riemann integral on Ω.

Double integral variable substitution:

$$\iint_D f(x,y)\, dx\, dy = \iint_{Duv} f(x(u,v), y(u,v)) \left| \frac{D(x,y)}{D(u,v)} \right| du\, dv$$

In polar coordinates:

$$\begin{cases} x = \rho\cos\varphi \\ y = \rho\sin\varphi \end{cases} (\rho \geq 0, 0 \leq \varphi < 2\pi) \quad d\sigma = dx\, dy = \rho\, d\rho\, d\varphi$$

Triple integral by cylindrical coordinates:

$$M(\Omega) = \iiint_\Omega \rho(x,y,z)\, dV = \int_a^b dx \int_{y_1(x)}^{y_2(x)} dy \int_{z_1(x,y)}^{z_2(x,y,z)} \rho(x,y,z)\, dz$$

Triple integral variable substitution:

$$\iiint_\Omega f(x,y,z)\, dV = \iiint_\Omega f(x(r,s,t), y(r,s,t), z(r,s,t)) \left| \frac{D(x,y,z)}{D(r,s,t)} \right| dr\, ds\, dt$$

In cylindrical coordinates:

$$\begin{cases} x = \rho\cos\varphi \\ y = \rho\sin\varphi \\ z = z \end{cases} \quad \frac{D(x,y,z)}{D(\rho,\varphi,z)} = \rho$$

In spherical coordinates:

$$\begin{cases} x = r\sin\varphi\cos\theta \\ y = r\sin\varphi\sin\theta \\ z = r\cos\varphi \end{cases} \quad \frac{D(x,y,z)}{D(r,\varphi,\theta)} = -r^2\sin\varphi \quad \begin{cases} r \geq 0 \\ 0 \leq \varphi < \pi \\ 0 \leq \theta < 2\pi \end{cases}$$

Example: Volume of a unit sphere in Rn:

$$V_n = \int_{\Omega_n} dV = \int_{-1}^{1} dx_n \int_{-1}^{1} \ldots \int_{-1}^{1} dx_1 \ldots dx_{n-1} \quad \Omega_{n-1} = \{(x_1, \ldots, x_{n-1}) \mid x_1^2 + \ldots + x_{n-1}^2 \leq 1 - x_n^2\}$$

$$\therefore V_n = \int_{\Omega_n} (\sqrt{1 - x_n^2})^{n-1} \, dx_n \int_{-1}^{1} \ldots \int_{-1}^{1} du_1 \ldots du_{n-1}$$

$$\therefore \frac{D(x_1, \ldots, x_{n-1})}{D(u_1, \ldots, u_{n-1})} = (\sqrt{1 - x_n^2})^{n-1} \qquad \therefore V_n = \int_{\Omega_n} (\sqrt{1 - x_n^2})^{n-1} \, dx_n \int_{-1}^{1} \ldots \int_{-1}^{1} du_1 \ldots du_{n-1}$$

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$$V_n = V_{n-1}\int_{-1}^{1}(\sqrt{1-t^2})^n dt = V_{n-1}\cdot B\left(\frac{1}{2},\frac{n+1}{2}\right)$$

$$\therefore V_n = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}\sqrt{\pi}V_{n-1}$$

$$\therefore V_{2m} = \frac{\pi^m}{m!}, \quad V_{2m+1} = \frac{2^m\pi^{m-1}}{(2m-1)!}$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_2 & y_2 & 1

\end{vmatrix} = A$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_3 & y_3 & 1

\end{vmatrix} = B$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_4 & y_4 & 1

\end{vmatrix} = C$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_5 & y_5 & 1

\end{vmatrix} = D$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_6 & y_6 & 1

\end{vmatrix} = E$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_7 & y_7 & 1

\end{vmatrix} = F$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_8 & y_8 & 1

\end{vmatrix} = G$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_9 & y_9 & 1

\end{vmatrix} = H$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{10} & y_{10} & 1

\end{vmatrix} = I$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{11} & y_{11} & 1

\end{vmatrix} = J$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{12} & y_{12} & 1

\end{vmatrix} = K$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{13} & y_{13} & 1

\end{vmatrix} = L$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{14} & y_{14} & 1

\end{vmatrix} = M$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{15} & y_{15} & 1

\end{vmatrix} = N$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{16} & y_{16} & 1

\end{vmatrix} = O$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{17} & y_{17} & 1

\end{vmatrix} = P$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{18} & y_{18} & 1

\end{vmatrix} = Q$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{19} & y_{19} & 1

\end{vmatrix} = R$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{20} & y_{20} & 1

\end{vmatrix} = S$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{21} & y_{21} & 1

\end{vmatrix} = T$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{22} & y_{22} & 1

\end{vmatrix} = U$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{23} & y_{23} & 1

\end{vmatrix} = V$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{24} & y_{24} & 1

\end{vmatrix} = W$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{25} & y_{25} & 1

\end{vmatrix} = X$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{26} & y_{26} & 1

\end{vmatrix} = Y$$

$$\begin{vmatrix}

x & y & 1 \

x_1 & y_1 & 1 \

x_{27} & y_{27} & 1

\end{vmatrix} = Z$$

$$\therefore A \cdot B \cdot C \cdot D \cdot E \cdot F \cdot G \cdot H \cdot I \cdot J \cdot K \cdot L \cdot M \cdot N \cdot O \cdot P \cdot Q \cdot R \cdot S \cdot T \cdot U \cdot V \

Chapter 4: First Kind of Line Integrals and First Kind of Surface Integrals

First Kind of Line Integrals:

$$\int_{AB} f(x,y,z)dl = \int_{BA} f(x,y,z)dl$$

$$\int_{L} f(x,y,z)dl = \int_{L_1} f(x,y,z)dl + \int_{L_2} f(x,y,z)dl$$

$$\int_{L} f(x,y,z)dl = \int_{a}^{b} f(x(t),y(t),z(t))\sqrt{x'(t)^2+y'(t)^2+z'(t)^2}dt$$

First Kind of Surface Integrals: If the surface S's equation is parametric form

$$\begin{cases} x = x(u,v) \\ y = y(u,v) \\ z = z(u,v) \end{cases}$$

Then the surface's normal vector is

Missing \end{pmatrix}

Then $|\cos(\vec{n},\vec{z})| = \frac{C}{\sqrt{A^2+B^2+C^2}}$, $ds = \frac{\partial x \partial y}{|\cos(\vec{n},\vec{z})|} = \frac{1}{\cos(\vec{n},\vec{z})} \cdot \frac{\partial(x,y)}{\partial(u,v)}dudv = \sqrt{A^2+B^2+C^2}dudv$

And $A^2+B^2+C^2 = EG-F^2$, where $E = 1+(\frac{\partial z}{\partial x})^2$, $G = 1+(\frac{\partial z}{\partial y})^2$, $F = \frac{\partial z}{\partial x}\frac{\partial z}{\partial y}$.

$\therefore ds = \sqrt{EG-F^2}dudv.$

$\therefore \iint_S f(x,y,z)ds = \iint_{D_{xy}} f(x(u,v),y(u,v),z(u,v))\sqrt{EG-F^2}dudv.$

Example:

$\iint_S f(x,y,z)ds = \iint_{D_{xy}} f(x,y,z(x,y))\sqrt{1+(\frac{\partial z}{\partial x})^2+(\frac{\partial z}{\partial y})^2}dxdy.$

Example:

The surface area element $ds = \sqrt{EG-F^2}d\varphi dv = r^2 \sin\varphi d\varphi dv.$

Chapter 5: Second Kind Line Integrals and Second Kind Surface Integrals

Second Kind Line Integral:

$$\int_{L(A)}^{(B)} \vec{F} \cdot d\vec{r} = -\int_{L(B)}^{(A)} \vec{F} \cdot d\vec{r}$$

Vector Form:

$$\int_{L(A)}^{(B)} \vec{F} \cdot d\vec{r} = \int X dx + Y dy + Z dz = \int_a^b X \cdot x'(t)dt + \int_a^b Y \cdot y'(t)dt + \int_a^b Z \cdot z'(t)dt$$

Converted to First Kind Line Integral:

$$\int_{L(A)}^{(B)} \vec{F} \cdot d\vec{r} = \int_{AB} (X \cos\alpha + Y \cos\beta + Z \cos\gamma) ds$$

Second Kind Surface Integral:

$$\iint_{S^+} \vec{F} \cdot d\vec{S} = \iint_{S^+} (X \cos\alpha + Y \cos\beta + Z \cos\gamma) ds$$

Properties:

$$\iint_{S^+} \vec{F} \cdot d\vec{S} = -\iint_{S^+} \vec{F} \cdot d\vec{S}$$

Parametric Form:

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v)) \quad (u, v) \in D_{uv}$$

When $\vec{r}_u \times \vec{r}_v \neq 0$, unit tangent vector $\vec{r}_s = \pm \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} = \pm \frac{(A,B,C)}{\sqrt{A^2+B^2+C^2}} = \pm(\cos\alpha, \cos\beta, \cos\gamma)$

And $ds = \sqrt{A^2 + B^2 + C^2}\, dudv$

Therefore:

$$dy \wedge dz = \cos\alpha\, ds = \pm A\, dudv$$

$$dz \wedge dx = \cos\beta\, ds = \pm B\, dudv$$

$$dx \wedge dy = \cos\gamma\, ds = \pm C\, dudv$$

$$\therefore \iint_{S^+} \vec{F} \cdot d\vec{S} = \pm \iint_{D_{uv}} (XA + YB + ZC)\, dudv$$

Green's Theorem:

$$\int_S \nabla \times \vec{V} \cdot d\vec{S} = \oint_{\partial S} \vec{V} \cdot d\vec{\ell}$$

where

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

If

$$\nabla \times \vec{V} = 0$$

, then the second kind of line integral

$$\int_{A_1}^{A_2} \vec{V} \cdot d\vec{\ell}$$

is path-independent. And

$$\int_{A_1}^{A_2} \vec{V} \cdot d\vec{\ell} = u(X)\Big|_{A_1}^{A_2} = u(A_2) - u(A_1)$$

: primitive function.

Gauss's Theorem:

$$\int_\Omega \nabla \cdot \vec{F} \cdot dV = \oint_{\partial\Omega} \vec{F} \cdot d\vec{S}$$

where

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Theorem: In a simply connected region, a vector field

$$\vec{V} = (X, Y, Z)$$

is a gradient field if and only if there exists a function

$$u(x, y, z)$$

such that

$$du = Xdx + Ydy + Zdz$$

(

$$R.P. X = \frac{\partial u}{\partial x}, Y = \frac{\partial u}{\partial y}, Z = \frac{\partial u}{\partial z}, du = \nabla u \cdot d\vec{r}$$

).

$$u$$

is the potential function of

$$\vec{V}$$

.

$$\nabla = \hat{e_x}\frac{\partial}{\partial x} + \hat{e_y}\frac{\partial}{\partial y} + \hat{e_z}\frac{\partial}{\partial z}$$

$$\nabla u \cdot \vec{f} = \frac{\partial}{\partial x} f_x + \frac{\partial}{\partial y} f_y + \frac{\partial}{\partial z} f_z$$

$$\mathrm{rot}\vec{f} = \begin{vmatrix} \hat{e_x} & \hat{e_y} & \hat{e_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_x & f_y & f_z \end{vmatrix}$$

$$\mathrm{grad}\varphi = \frac{\partial \varphi}{\partial x}\hat{e_x} + \frac{\partial \varphi}{\partial y}\hat{e_y} + \frac{\partial \varphi}{\partial z}\hat{e_z}$$

Theorem:

$$\nabla \times \nabla \varphi = 0 \Rightarrow$$

a solenoidal field can be expressed as the gradient of a scalar potential. (Continuous and differentiable).

$$\nabla \cdot \nabla \times \vec{f} = 0 \Rightarrow$$

a solenoidal field can be expressed as the curl of another vector field.

Chapter 6: Space Curves and Space Surfaces

In R^3, a vector function is defined as:

$$\vec{r}(t) = (x(t), y(t), z(t)), t \in [a, b]$$

The derivative at t0:

$$d\vec{r}(t_0) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)^T_{t_0} dt$$

The derivative at t0:

$$\vec{r}'(t_0) = (x'(t_0), y'(t_0), z'(t_0)) : R^1 \to R^3$$

Properties:

$$\frac{d}{dt}(\vec{r}_1 \times \vec{r}_2) = \frac{d\vec{r}_1}{dt} \times \vec{r}_2 + \vec{r}_1 \times \frac{d\vec{r}_2}{dt}$$

$$\frac{d}{dt}(\vec{r}_1 \cdot \vec{r}_2 \cdot \vec{r}_3) = (\frac{d\vec{r}_1}{dt}, \vec{r}_2, \vec{r}_3) + (\vec{r}_1, \frac{d\vec{r}_2}{dt}, \vec{r}_3) + (\vec{r}_1, \vec{r}_2, \frac{d\vec{r}_3}{dt})$$

where

$$(\vec{r}_1, \vec{r}_2, \vec{r}_3) = \vec{r}_1 \cdot (\vec{r}_2 \times \vec{r}_3)$$

If $\vec{F}(t)$ is n times continuously differentiable, then each component of $\vec{F}(t)$ is n times continuously differentiable.

Vector function integration:

$$\int \vec{F}(t) dt = \vec{F}(t) + \vec{C}$$

Properties:

$$\int \vec{a} \cdot \vec{F}(t) dt = \vec{a} \cdot \int \vec{r}(t) dt$$

$$\int \vec{a} \times \vec{F}(t) dt = \vec{a} \times \int \vec{F}(t) dt$$

Arc length:

$$s = \int_a^b ||\vec{r}(t)||dt = \int_a^b \sqrt{x^2(t) + y^2(t) + z^2(t)}dt$$

Since $ds > 0$, the inverse function $t = t(s)$ exists, allowing us to parameterize the curve by arc length:

$$\vec{F} = \vec{F}(s) = (x(s), y(s), z(s))$$

Convention:

$$\vec{F} = \frac{d\vec{F}}{dt}, \vec{F}' = \frac{d\vec{F}}{ds}$$

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The tangent vector of the curve: $\vec{T} = \vec{r}'(s_0)$, $\|\vec{T}\| = 1$

The tangent line equation: $\vec{p} = \vec{r}(s_0) + \lambda\vec{T} = \vec{r}(s_0) + \lambda\vec{r}'(s_0)$. General parameter: $\vec{T} = \frac{\vec{r}(t)}{\|\vec{r}(t)\|}$

The tangent plane equation: $(\vec{p} - \vec{r}(s_0)) \cdot \vec{T} = (\vec{p} - \vec{r}(s_0)) \cdot \vec{r}'(s_0) = 0$.

The unit normal vector: $\vec{B} = \frac{\vec{r}'(s_0)\times\vec{r}''(s_0)}{\|\vec{r}'(s_0)\times\vec{r}''(s_0)\|}$

The normal line equation: $\vec{p} = \vec{r}(s_0) + \lambda\vec{B}$ General parameter: $\vec{B} = \frac{\vec{r}'(t)\times\vec{r}''(t)}{\|\vec{r}'(t)\times\vec{r}''(t)\|}$

The tangent plane equation: $(\vec{p} - \vec{r}(s_0)) \cdot \vec{B} = 0$.

The unit binormal vector: $\vec{N} = \frac{\vec{r}'''(s_0)}{\|\vec{r}'''(s_0)\|}$

The binormal line equation: $\vec{p} = \vec{r}(s_0) + \lambda\vec{N}$ General parameter: $\vec{N} = \frac{(\vec{r}\times\vec{r}')\times\vec{r}'}{\|\vec{r}\times\vec{r}'\|\|\vec{r}'\|}$

The tangent plane equation: $(\vec{p} - \vec{r}(s_0)) \cdot \vec{N} = 0$.

Frenet's Frame (natural frame): $\{\vec{F}(s), \vec{T}(s), \vec{N}(s), \vec{B}(s)\}$

Curvature: $k(s) = \|\vec{r}'(s)\|$ General parameter: $k(t) = \frac{\|\vec{r}'(t)\times\vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$

Curvature radius: $R(s) = \frac{1}{k(s)}$. $\vec{T}'(s) = k(s)\vec{N}(s)$

Curvature center: $\vec{r}(s) + R(s)\vec{N}(s)$ $k(s) = \frac{d\varphi}{ds}$.

Torsion: $\tau(s) = -\vec{B}(s) \cdot \vec{N}(s)$. $|\tau(s)| = \left|\frac{d\varphi}{ds}\right|$

General parameter: $\tau = \frac{\left(\vec{F}, \vec{F}', \vec{F}''\right)}{\left(\vec{F}\times\vec{F}'\right)^2}$

$\vec{B}' = -\tau\vec{N}$

Frenet's Formula: $\begin{pmatrix} \vec{T}' \\ \vec{N}' \\ \vec{B}' \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} \vec{T} \\ \vec{N} \\ \vec{B} \end{pmatrix}$

The arc length parameter s, curvature k(s), and torsion τ(s) are the basic quantities describing a curve.

The parametric form of a surface is: $\vec{r} = \vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$

u curve: Fix v and vary u.

v curve: Fix u and vary v.

Parameter transformation: $\vec{r}_u \times \vec{r}_v = \frac{D(u,v)}{|\vec{r}_u \times \vec{r}_v|}(\vec{r}_u \times \vec{r}_v)$. By the chain rule of composite functions.

$\vec{r}_u, \vec{r}_v$: Basis of the tangent space.

Normal vector: $\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$

Tangent plane: $\vec{n} \cdot (\vec{p} - \vec{r}) = 0$. Parametric equation: $\vec{p} = \vec{r} + \lambda \vec{r}_u + \mu \vec{r}_v$.

Normal equation: $\vec{p} = \vec{r} + \lambda \vec{n}$

The first fundamental form of the surface:
$ds^2 = d\vec{r} \cdot d\vec{r} = (\vec{r}_u du + \vec{r}_v dv)^2 = \vec{r}_u^2 du^2 + 2\vec{r}_u \cdot \vec{r}_v du dv + \vec{r}_v^2 dv^2$

$= (du, dv) \begin{bmatrix} E & F \\ F & G \end{bmatrix} \begin{bmatrix} du \\ dv \end{bmatrix}$.

The first fundamental form is independent of the parameter transformation.

The curvature of a curve on the surface: $k\vec{N} = k_n \vec{n} + k_g(\vec{n} \times \vec{r})$. k_n is the normal curvature, k_g is the geodesic curvature.

k: curvature

$\therefore k_n = k\vec{N} \cdot \vec{n} = \vec{r}' \cdot \vec{n} = \vec{r}_u \cdot \vec{n}$

$\begin{aligned}

\angle &= -\vec{r}_{u} \cdot \vec{r}_{u} \

M &= -\vec{r}_{u} \cdot \vec{r}_{v} = -\vec{r}_{v} \cdot \vec{r}_{u} \

N &= -\vec{r}_{v} \cdot \vec{r}_{v}

\end{aligned}$

$\begin{aligned}

k_n &= -\frac{d\vec{r} \cdot d\vec{r}_{u}}{ds^2} \

&= \frac{(\vec{r}_{u} u + \vec{r}_{v} v)'}{\vec{r}_{uu} \cdot \vec{n}} \cdot \vec{n} \

&= (\vec{r}_{uu} \cdot \vec{n}) u^2 + 2 \vec{r}_{uv} \cdot \vec{n} u v + (\vec{r}_{vv} \cdot \vec{n}) v^2 \

&= \frac{du^2 + 2M du dv + N dv^2}{F} \

&= \frac{du}{dv}

$$\begin{bmatrix} \frac{2}{F} & M \\ M & N \end{bmatrix}$$

\frac{du}{dv}

\end{aligned}$